

counter.svelte: Counter component

Project reference: STN22_Thoracic belt for respiratory rate measurement

Authors: Hakkou Dounia

Date: 2025–2026 © SIS TEAM NANCY

1. Source code

```
<!--
  Description: Reactive counter component for Svelte.
  Declares a reactive numeric variable initialized to 0.
  Renders a button displaying the current count value.
  Each click triggers the increment() function,
  which increases count by 1 and automatically re-renders the UI.

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-->

<script lang="ts">
  // Reactive variable 'count' typed as number, initialized to 0.
  // In Svelte, any 'let' variable at the top level is reactive:
  // the DOM updates automatically whenever its value changes.
  let count: number = 0

  // Arrow function that increments count by 1 on each call.
  // Svelte detects the assignment (count += 1) and triggers
  // a reactive re-render of every element that uses count.
  const increment = () => {
    count += 1
  }
</script>

<!-- Button bound to the increment function via the on:click directive. -->
<!-- Svelte's on:click is equivalent to addEventListener('click', ...). -->
<button on:click={increment}>
  <!-- Svelte interpolation: {count} is replaced by the current value. -->
  <!-- The text updates instantly in the DOM on each state change. -->
  count is {count}
</button>
```